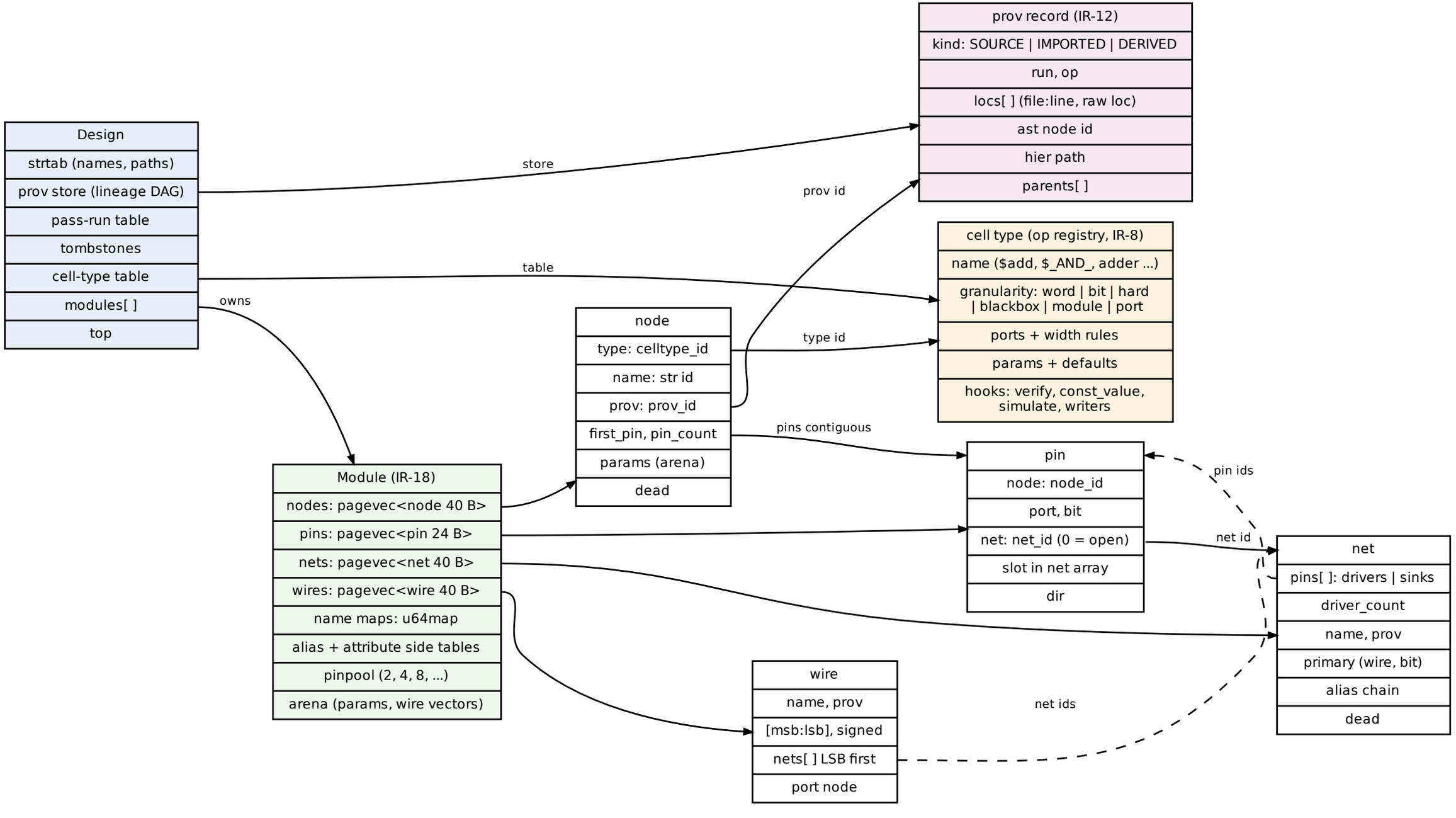
Design
strtab (names, paths)
prov store (lineage DAG)
pass-run table
tombstones
cell-type table
modules[ ]
top
store
prov record (IR-12)
kind: SOURCE | IMPORTED | DERIVED
run, op
locs[ ] (file:line, raw loc)
ast node id
hier path
parents[ ]
prov id
table
cell type (op registry, IR-8)
name ($add, $_AND_, adder ...)
granularity: word | bit | hard | blackbox | module | port
ports + width rules
params + defaults
hooks: verify, const_value, simulate, writers
owns
node
type: celltype_id
name: str id
prov: prov_id
first_pin, pin_count
params (arena)
dead
type id
pins contiguous
Module (IR-18)
nodes: pagevec<node 40 B>
pins: pagevec<pin 24 B>
nets: pagevec<net 40 B>
wires: pagevec<wire 40 B>
name maps: u64map
alias + attribute side tables
pinpool (2, 4, 8, ...)
arena (params, wire vectors)
pin
node: node_id
port, bit
net: net_id (0 = open)
slot in net array
dir
pin ids
net id
net
pins[ ]: drivers | sinks
driver_count
name, prov
primary (wire, bit)
alias chain
dead
wire
name, prov
[msb:lsb], signed
nets[ ] LSB first
port node
net ids